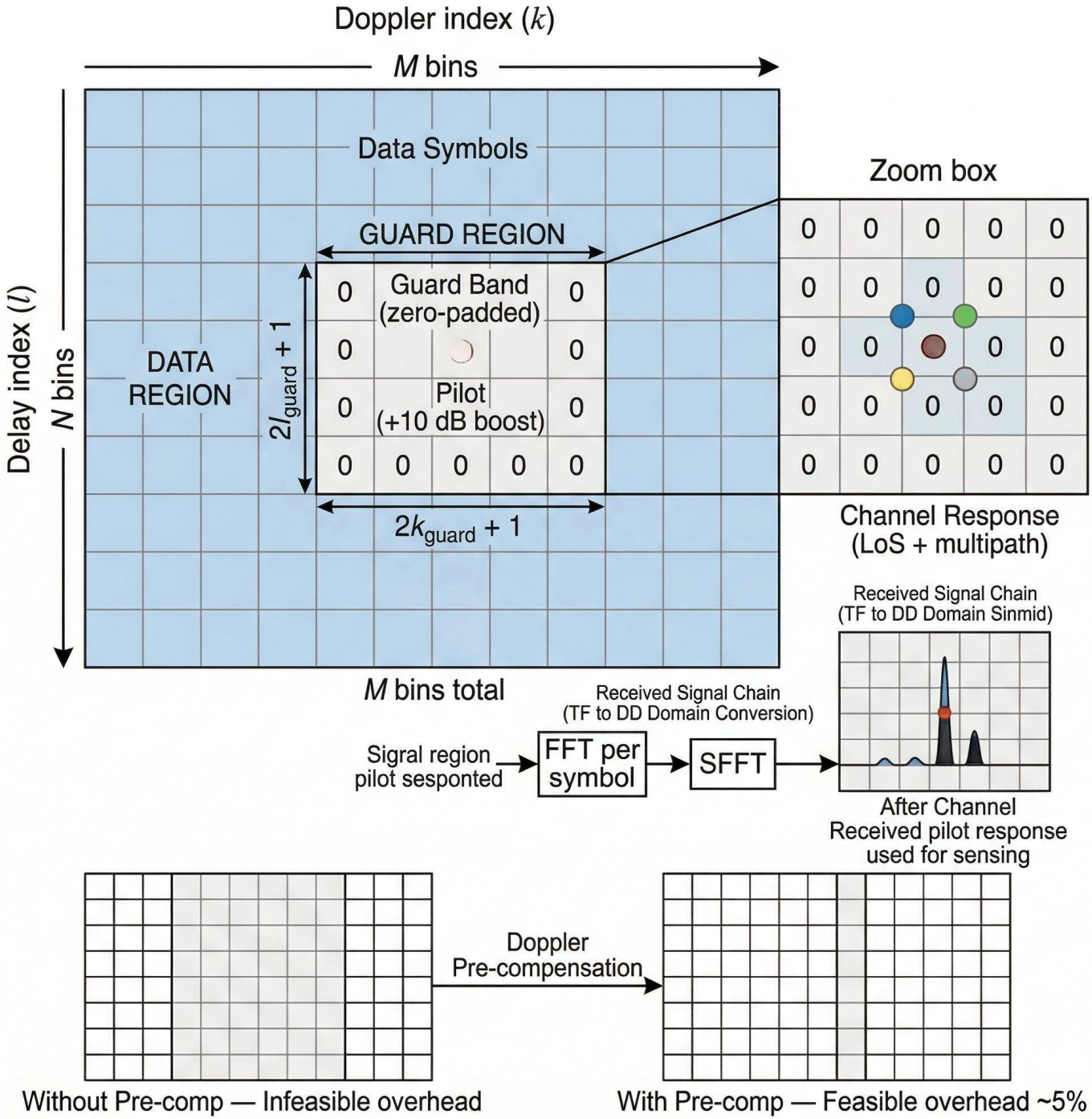
Doppler index (k)
M bins
Data Symbols
GUARD REGION
Guard Band
(zero-padded)
Pilot
(+10 dB boost)
DATA
REGION
$2l_{guard} + 1$
$2k_{guard} + 1$
Delay index (l)
N bins
M bins total
Zoom box
Channel Response
(LoS + multipath)
Received Signal Chain
(TF to DD Domain Sinmid)
Received Signal Chain
(TF to DD Domain Conversion)
Sigral region
pilot sesponted
FFT per
symbol
SFFT
After Channel
Received pilot response
used for sensing
Doppler
Pre-compensation
Without Pre-comp — Infeasible overhead
With Pre-comp — Feasible overhead ~5%